

Communication Diagram

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title Communication Diagram: Membuat Purchase Order

skinparam linestyle ortho

skinparam shadowing false

actor "Staff Gudang" as Actor

rectangle "UI_PO" as UI

rectangle "ProcurementHandler" as Ctrl

rectangle "MS_Supplier" as Supplier

rectangle "MS_Warehouse" as Warehouse

rectangle "TR_Procurement" as POHeader

rectangle "TR_DetailProcurement" as POItem

rectangle "MS_Material" as Material

' Messages

Actor → UI : 1: inputPO(data)

UI → Ctrl : 1.1: createDraftPO(data)

' Validasi Data Pendukung

Ctrl → Supplier : 1.1.1: getSupplierDetails()

Supplier → Warehouse : 1.1.2: checkStockLevel()

Warehouse → Material : 1.1.3: getMaterialInfo()

' Create Header

Material → POHeader : 1.1.4: createHeaderPO()

```
' Loop Create Detail (Delegasi ke Header)
POHeader → POItem : 1.1.4.1: addItemToPO()
```

```
' Return Path
POHeader → Ctrl : 1.1.5: return(poID)
Ctrl → UI : 1.2: return(success)
UI → Actor : 1.3: displaySuccess()
```

```
note bottom of POHeader
    1.1.4.1 dilakukan
    berulang (Loop)
end note
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```

```
@startuml
title Communication Diagram: Manage Laporan Pemeriksaan Kualitas
skinparam linestyle ortho
skinparam shadowing false
```

```
actor "Staff QC" as Actor
rectangle ":UI_Inspection" as UI
rectangle ":MaterialInspectionHandler" as Ctrl
rectangle ":TR_Procurement" as PO
rectangle ":TR_PurchaseComplianceInspection" as PCI
rectangle ":TR_DetailMaterialInspection" as Detail
```

```
' Messages
Actor → UI : 1: submitResult(data)
UI → Ctrl : 1.1: submitInspectionResult(data)
```

```
' Get Reference
```

```

Ctrl → PO : 1.1.1: getPOData()

' Create Header QC
PO → PCI : 1.1.2: createInspectionRecord()

' Add Details (Delegasi)
PCI → Detail : 1.1.2.1: addDetailResult()

' Return Path
PCI → Ctrl : 1.1.3: return(idInspection)
Ctrl → UI : 1.2: showSuccess()
UI → Actor : 1.3: display()

note bottom of PCI
    1.1.2.1 Loop untuk
    setiap material
end note
@enduml

```

```

@startuml
title Communication Diagram: Manage Jadwal Pengiriman
skinparam linestyle ortho
skinparam shadowing false

```

```

actor "Staff Gudang" as Actor
rectangle ":UI_Shipping" as UI
rectangle ":ShippingHandler" as Ctrl
rectangle ":MS_Vehicle" as Vehicle
rectangle ":TR_Delivery" as Delivery
rectangle ":TR_Order" as Order
rectangle ":MS_MaterialWarehouse" as Stock

```

' Messages

Actor → UI : 1: createSchedule()

UI → Ctrl : 1.1: scheduleShipment()

' Cek Kendaraan

Ctrl → Vehicle : 1.1.1: checkAvailability()

' Create Pengiriman

Vehicle → Delivery : 1.1.2: createDeliverySchedule()

' Update Entitas Lain via Delivery (Delegasi)

Delivery → Order : 1.1.2.1: updateStatus("Shipping")

Order → Stock : 1.1.2.2: decreaseStock()

' Return Path

Delivery → Ctrl : 1.1.3: return(deliveryID)

Ctrl → UI : 1.2: showSuratJalan()

UI → Actor : 1.3: print()

note bottom of Delivery

Delivery mengorkestrasi

update stok & order

end note

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title Communication Diagram: Membuat Laporan Pengambilan Bahan

skinparam linestyle ortho

skinparam shadowing false

actor "Tim Produksi" as Actor
rectangle ":UI_Intake" as UI
rectangle ":MaterialIntakeHandler" as Ctrl
rectangle ":TR_ProductionPlanning" as Plan
rectangle ":TR_DailyMaterialIntake" as DMI
rectangle ":TR_DetailMaterialIntake" as Detail
rectangle ":MS_MaterialWarehouse" as Stock

' Messages

Actor → UI : 1: inputIntake(data)

UI → Ctrl : 1.1: recordIntake(data)

' Validasi Job Order

Ctrl → Plan : 1.1.1: validateJobOrder()

' Create Header Intake

Plan → DMI : 1.1.2: createIntakeRecord()

' Create Detail (Loop)

DMI → Detail : 1.1.2.1: addDetailItem()

' Update Stock fisik saat item ditambah

Detail → Stock : 1.1.2.1.1: decreaseStock()

' Return Path

Detail → DMI : 1.1.2.2: return

```
DMI → Ctrl : 1.1.3: return(success)
Ctrl → UI : 1.2: displaySuccess()
UI → Actor : 1.3: finish()
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```

```
@startuml
title Communication Diagram: Manage Laporan Hasil QC
skinparam linestyle ortho
skinparam shadowing false
```

```
actor "Staff QC" as Actor
rectangle ":UI_QCResult" as UI
rectangle ":ProductQCHandler" as Ctrl
rectangle ":TR_ProductionResult" as Result
rectangle ":TR_ProductAnalysis" as Chem
rectangle ":TR_Microbiology" as Micro
rectangle ":MS_MaterialWarehouse" as Stock
```

```
' Messages
```

```
Actor → UI : 1: submitLabData()
UI → Ctrl : 1.1: submitLabResult()
```

```
' Get Parent Data
```

```
Ctrl → Result : 1.1.1: getResultData()
```

```
' Save QC Data (Delegasi: Result mencatat QC-nya)
```

```
Result → Chem : 1.1.2: saveChemicalAnalysis()
```

```
Chem → Micro : 1.1.3: saveMicroResult()
```

```
' Update Status Stok (Triggered by Result)
Micro → Stock : 1.1.4: updateStockStatus("Released")
```

```
' Return Path
Result → Ctrl : 1.1.5: return(status)
Ctrl → UI : 1.2: showConfirmation()
UI → Actor : 1.3: display()
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```

```
@startuml
title Communication Diagram: Membuat Plan Produksi
skinparam linestyle ortho
skinparam shadowing false
```

```
actor "Tim Produksi" as Actor
rectangle ":UI_Planning" as UI
rectangle ":ProductionPlanningHandler" as Ctrl
rectangle ":MS_Product" as Product
rectangle ":MS_Recipe" as Recipe
rectangle ":MS_MaterialWarehouse" as Stock
rectangle ":TR_ProductionPlanning" as PP
rectangle ":TR_DetailProductionPlanning" as Detail
```

```
' Messages
Actor → UI : 1: createPlan(data)
UI → Ctrl : 1.1: createJobOrder(data)
```

' Cek BOM (Product tanya ke Recipe)
Ctrl → Product : 1.1.1: getProductDetails()
Product → Recipe : 1.1.1.1: calculateMaterial()

' Cek Stok
Product → Stock : 1.1.2: checkStockAvailability()

' Create Plan Header
Stock → PP : 1.1.3: createHeader()

' Add Detail Plan
PP → Detail : 1.1.3.1: addDetailPlan()

' Return Path
PP → Ctrl : 1.1.4: return(jolD)
Ctrl → UI : 1.2: showJobOrder()
UI → Actor : 1.3: printJO()
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